

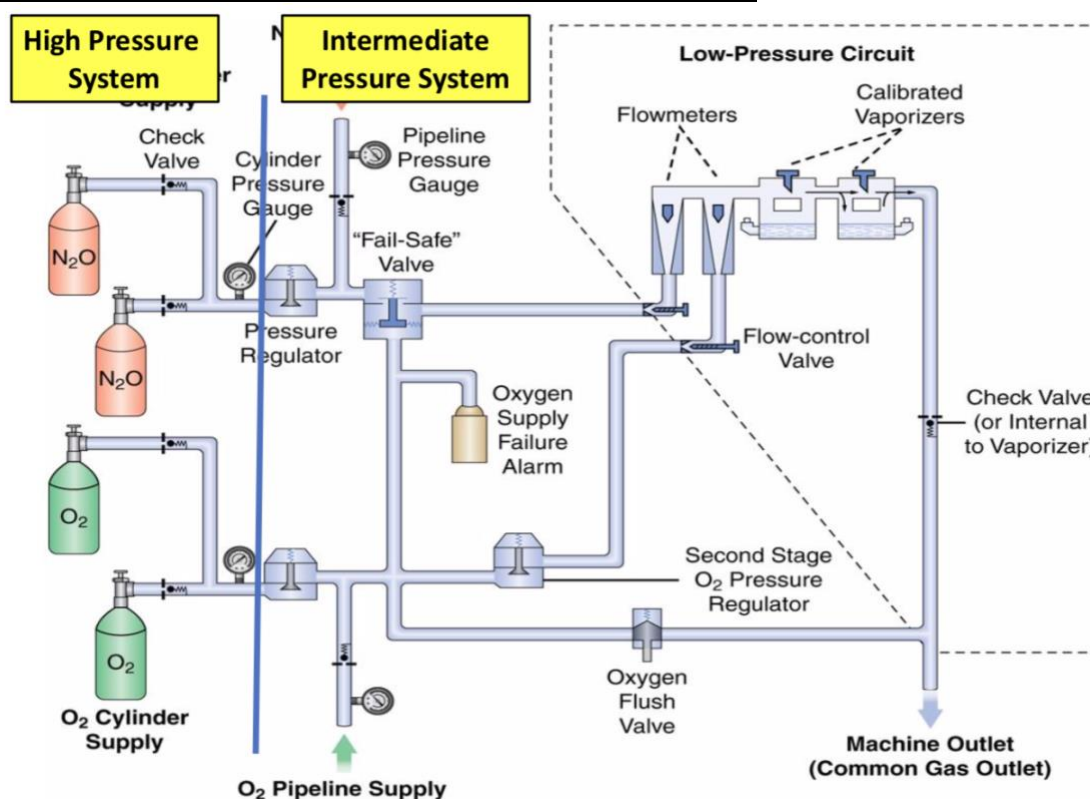
4. Describe the safety features of anaesthetic workstation in preventing delivery of hypoxic mixture.
(Modified from FHKCA 2017B)

Examiner Report

Describe the safety features of anaesthetic workstation in preventing delivery of hypoxic mixture. (5/23 passed)

This is a question of describing the safety features of anaesthetic workstation in preventing delivery of hypoxic mixture. It is considered basic to anaesthetic practice. Answers were non-specific and full of erroneous descriptions. For example, anti-hypoxic devices was described as "increase O₂ concentration when N₂O is increased, and decrease N₂O when reducing O₂ concentration (or similar descriptions)" but in fact it protects against an operator selected delivery of mixture of Oxygen and Nitrous oxide having an oxygen concentration below 21% Oxygen (v/v%) in the fresh gas or inspired gas. Oxygen supply failure alarm was activated because of "O₂ level (concentration in some candidates) significant drop" but this should be the drop in oxygen pressure below a manufacturer-specified minimum. No candidate mentioned it is not able to be muted until pressure is restored above that minimum level. A few candidates have difficulty in mentioning more than 5 features. Overall the answers were average with few deficient candidates mentioning 3-4 features only without description.

Reference Diagram (You do not need to replicate this for exams)



Sample Answer

1. **High pressure system 137 Bar** (Consists of Cylinder, 1st stage pressure regulator)

Cylinders

- Serves as O₂ reserve to pipeline supply (with preferential usage of pipeline > cylinder)
- Colour-coded, labelled for easy identification and to avoid misconnections.
- Pin Index Safety System (PISS) to ensure correct O₂ connection to anaesthesia machine
- Bodok seal to ensure tight connections, and prevent gas leak.
- Display of cylinder pressures to alert users to replace cylinders are exhausted

2. **Intermediate pressure system 4 Bar** (Consists of Pipeline, Fail-safe valve, O₂ supply failure alarm, Oxygen flush valve, 2nd stage pressure regulator)

Pipeline

- Colour-coded, labelled for easy identification and to avoid misconnections.
- Diameter Index Safety System (DISS) and unique Schrader probe profile to ensure correct O₂ connection at wall socket
- Non-interchangeable Screw Thread (NIST) for hose connection to anaesthesia machine
- Display of pipeline pressures to alert users if pressures are low

Fail-safe valve/ Hypoxic mixture prevention device

- Cessation of N₂O flow if no O₂ flow is present

O₂ supply failure alarm

- Generates an alarm and stops supply of other gases once O₂ pressure falls below a predetermined critical level. (2 Bar) (Ritchie Whistle =>8s. New alarms continues until O₂ pressure is restored)

Oxygen flush valve

- Delivers O₂ at 35-70 L/min, 4 Bar to facilitate rapid administration of O₂

3. **Low pressure system 1 Bar** (Flow control valve, flowmeters, vaporizers, check valve, common gas outlet and breathing circuit)

Flowmeter

- Large, fluted O₂ knob, located first from left on rotameter for easy identification of O₂ knob
- Minimum O₂ flow rate of 250ml/min (Equivalent to O₂ consumption of a resting adult)
- Anti-static material used for flowmeter to prevent stuck bobbin from impeding gas flows
- O₂ being the last gas to enter the common gas outlet to avoid administration of other gases should there be upstream leakage/cracks.
- Low O₂ delivery alarm (fuel cell analyser)
- Hypoxia Guard Proportioning system. Maintains ratio of O₂:N₂O at minimum ratio of 1:2

Breathing circuit

- Unidirectional valve to prevent mixture of expired gases (with lower FiO₂)
- CO₂ absorber to increase percentage of FiO₂ by reducing percentage of FiCO₂
- Reservoir bag to serve as FGF (and O₂) reservoir
- Paramagnetic O₂ analyser + Low O₂ delivery alarm which monitors FiO₂/ETO₂ (via sampling tube from breathing circuit)
- Standardised 22mm/15mm common gas outlet diameters to minimize risk of wrong connections

Patient Monitor

- Low SpO₂ alarm for warning of patient's hypoxemia